

Zachary Soriano

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EDUCATION

University of California, Los Angeles

Los Angeles, CA

Bachelor of Science, Mathematics of Computation (Math + CS)

Jun. 2028

- **GPA:** 3.68/4.00
- **Coursework:** Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Differential Equations

PROJECTS

Career Intel | Python, FastAPI, React.js, PostgreSQL, Claude API

- Full stack career intelligence app supporting career pivot advice, employment outlook, salary comparisons, more
- Engineered PostgreSQL + pgvector database ingesting 1,000+ occupations from O*NET and BLS
- Implemented hybrid RAG pipeline combining natural language to SQL query + cosine similarity vector search
- Designed React frontend with dynamic visualizations relevant to the user's query intent

DistilBERT Salary Prediction | Python (Scikit-learn, PyTorch, Hugging Face, Pandas, NumPy)

- Developed and deployed a DistilBERT-based model to predict salaries from unstructured job descriptions
- Achieved $R^2 = 0.69$ and $MAE = 0.21$ (log-scale), corresponding to predictions within 24% of true salary on average, outperforming baseline model (BoW + Linear Regression) by 7%
- Developed an interactive web interface (Hugging Face Spaces) enabling real-time salary inference from raw text inputs

Sound Scout – Full Stack Music Artist Recommendation Engine | Python, JavaScript, HTML/CSS, Flask

- Built a full-stack music recommendation web app using Flask + JavaScript serving 350,000+ artists
- Engineered recommendation system combining TF-IDF embeddings, cosine similarity, and K-means clustering for retrieval and grouping of artists, averaging 60ms latency

EXPERIENCE

Incoming Business Analyst Intern

Jun. 2026

Capital One

McLean, VA

- Incoming Summer 2026

Machine Learning Researcher

Apr. 2026 - Present

DataRes @ UCLA

Los Angeles, CA

- Developed a Graph Neural Network (GNN) pipeline on Reddit interaction data to learn user and post representations for AI-driven tasks such as recommendation, link prediction, and community classification
- Engineered graph-based features from user–post–comment interactions & subreddit structure, implemented models using PyTorch Geometric to capture higher-order relational patterns beyond text-only baselines

Nuclear Operations Training Intern

Jun. 2025 - Sep. 2025

Pacific Gas & Electric (PG&E)

Avila Beach, CA

- Built Power BI dashboard to highlight due dates of SAPNs of Maintenance Training using power query and the SAP SQL database, saving ~10 minutes a day for summary of due notifications
- Automated a long-term schedule for Nuclear Operations Initial License Training (ILT) using Oracle Primavera P6 with 600+ classroom/simulator events saving ~30 minutes per week over an 18 month term

Skills & Interests

Skills/Relevant Software: Python, C++, JavaScript, HTML/CSS, SQL

Frameworks & Tools: React.js, FastAPI, Flask, PyTorch, Scikit-learn, Pandas, NumPy, PostgreSQL, Power BI, Git

Interests: NCAA MBB, Card Collecting, Sneakers, Golf, Board Games, Fantasy Sports